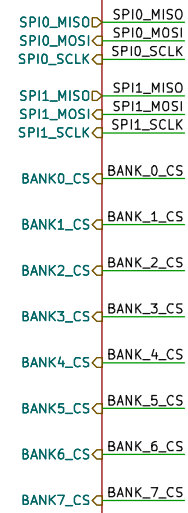
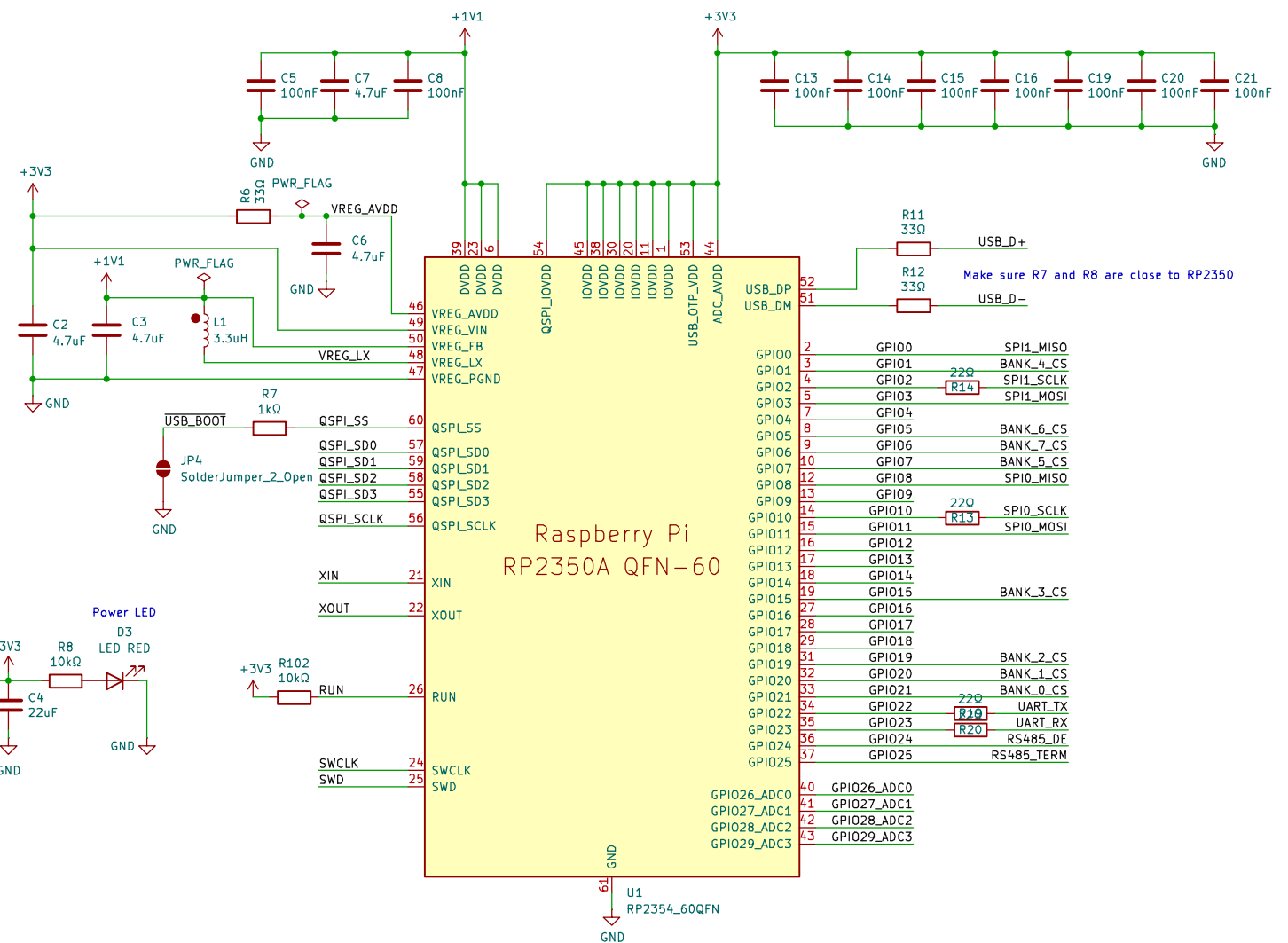


Sensor Array

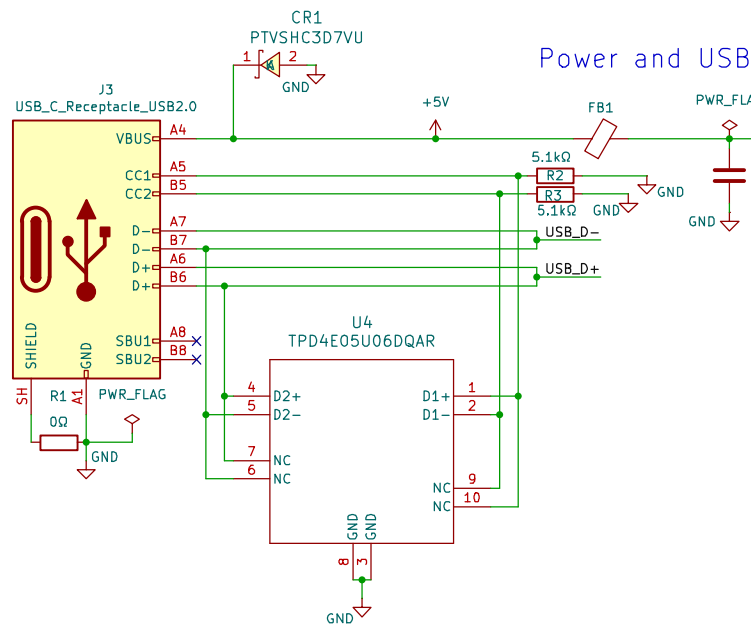
Sensor Array



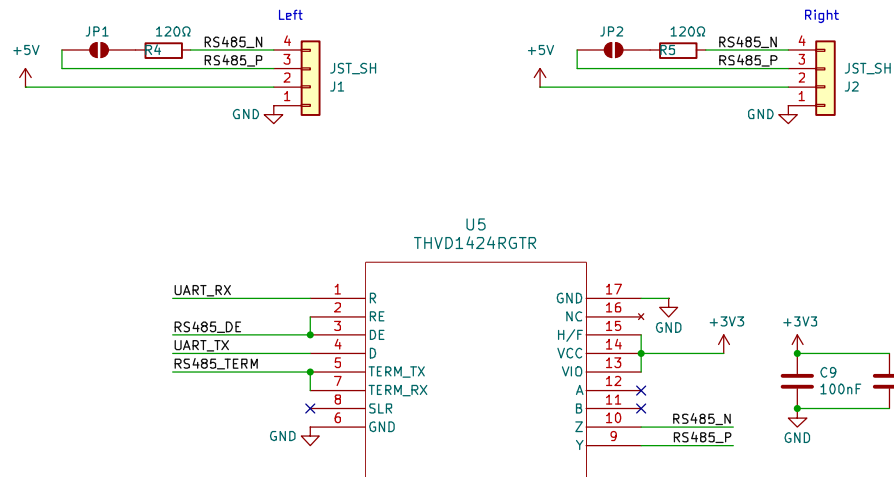
File: Sensor_Array.kicad_sch



Power and USB

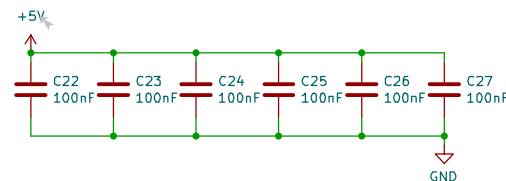
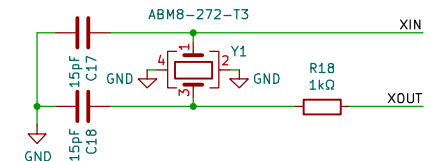


RS-485 Interface



This built-in RS485 termination should not be used.
Rather, terminate at the end of the board using the jumpers.

12MHz Crystal



5V RS485 VBUS
Noah Jaffe
Sheet: /
File: 002_Sensor_PL_RS485 board.kicad_sch
Title: PHOTON Sensor Board – A007
Size: A3 Date: 2026-04-22 Rev: B
KiCad E.D.A. 10.0.5 Id: 1/42

The (up-to) 32-sensor array consists of banks of 4.

Bank 0: Sensors 0–3

4_Sensor_bank

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO > SPI0_MISO
BANK_CS < BANK0_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank4

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO > SPI1_MISO
BANK_CS < BANK4_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank1

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO > SPI0_MISO
BANK_CS < BANK1_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank5

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO > SPI1_MISO
BANK_CS < BANK5_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank2

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO > SPI0_MISO
BANK_CS < BANK2_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank6

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO > SPI1_MISO
BANK_CS < BANK6_CS

File: 4_Sensor_Bank.kicad_sch

4_Sensor_bank3

SPI_SCLK < SPI0_SCLK
SPI_MOSI < SPI0_MOSI
SPI_MISO > SPI0_MISO
BANK_CS < BANK3_CS

File: 4_Sensor_Bank.kicad_sch

Last_sensor_bank

SPI_SCLK < SPI1_SCLK
SPI_MOSI < SPI1_MOSI
SPI_MISO > SPI1_MISO
BANK_CS < BANK7_CS

File: last_sensor_bank.kicad_sch

Note: This sensor bank does *not* inherit from 4_Sensor_bank
This is to allow for DNP sensors, so the total number of sensors
can be a number not modulus 4

SPI Sensor Bank Connections		
Noah Jaffe		
Sheet: /Sensor Array/ File: Sensor_Array.kicad_sch		
Title: 8x4-Channel LED Sensor Array		
Size: A4	Date: 2026-04-22	Rev: B
KiCad E.D.A. 10.0.5		Id: 2/42

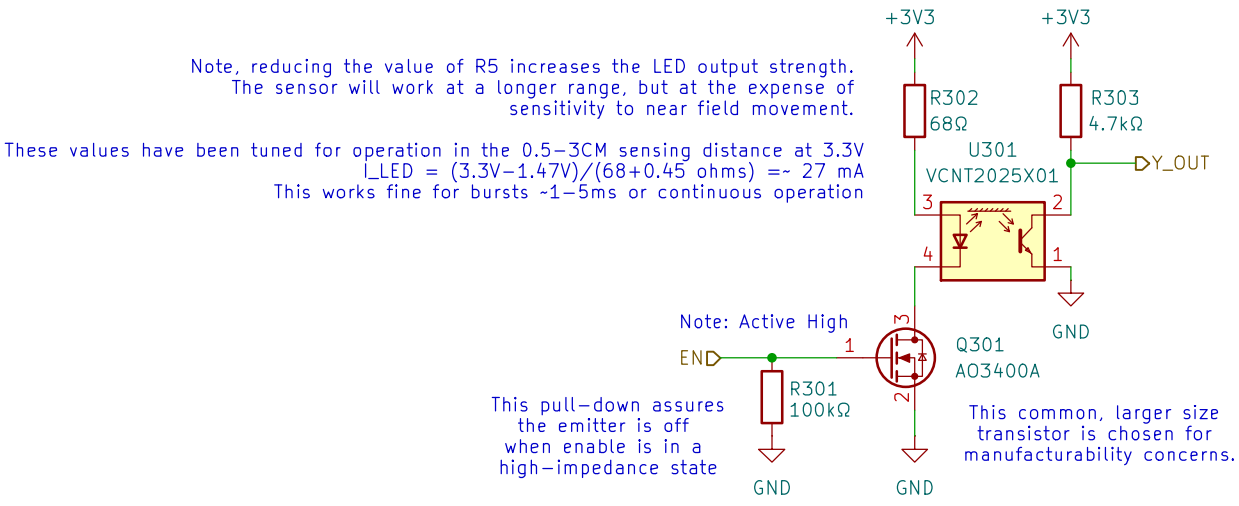
Noah Jaffe

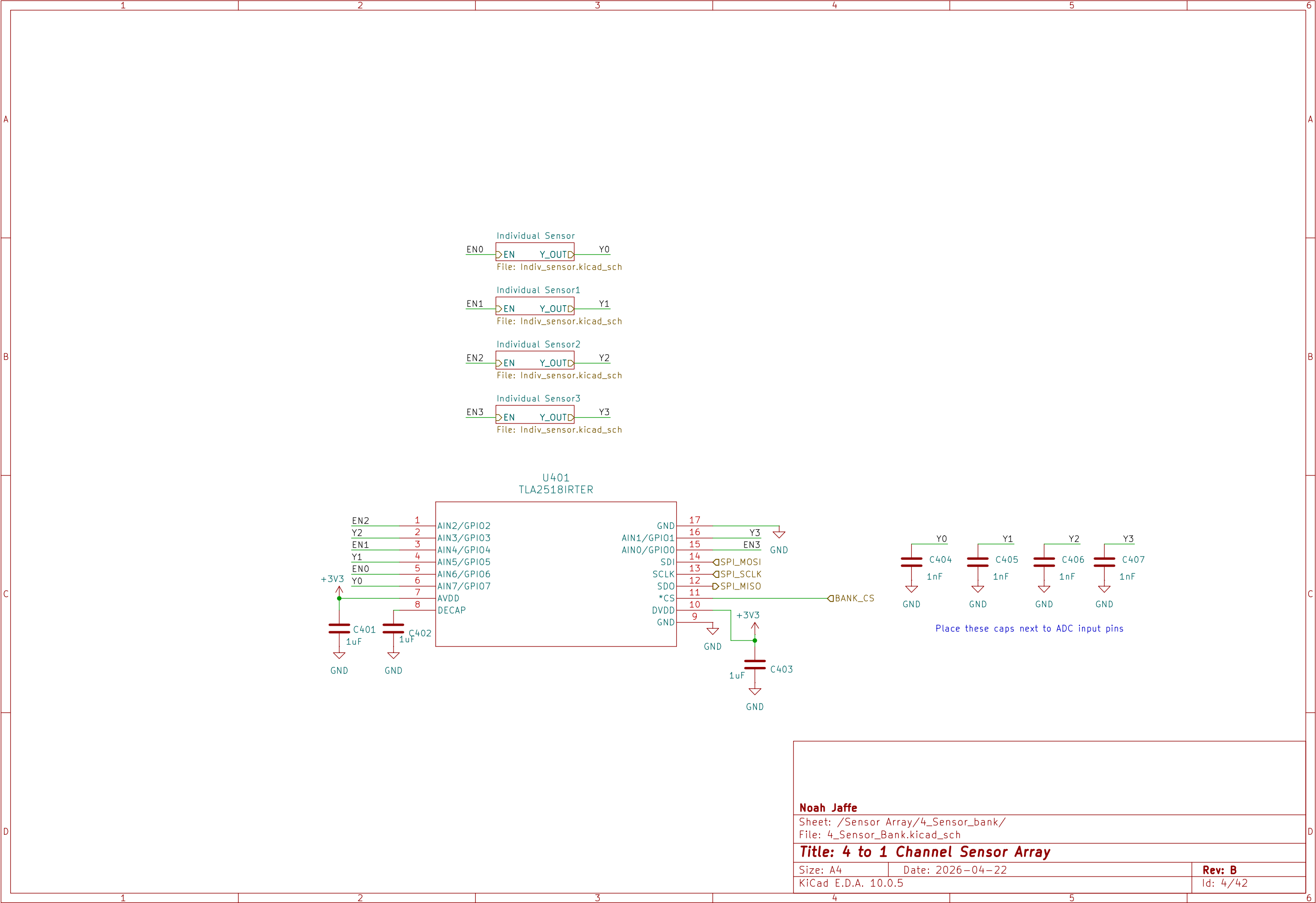
File: Sensor_Array.kicad_sch

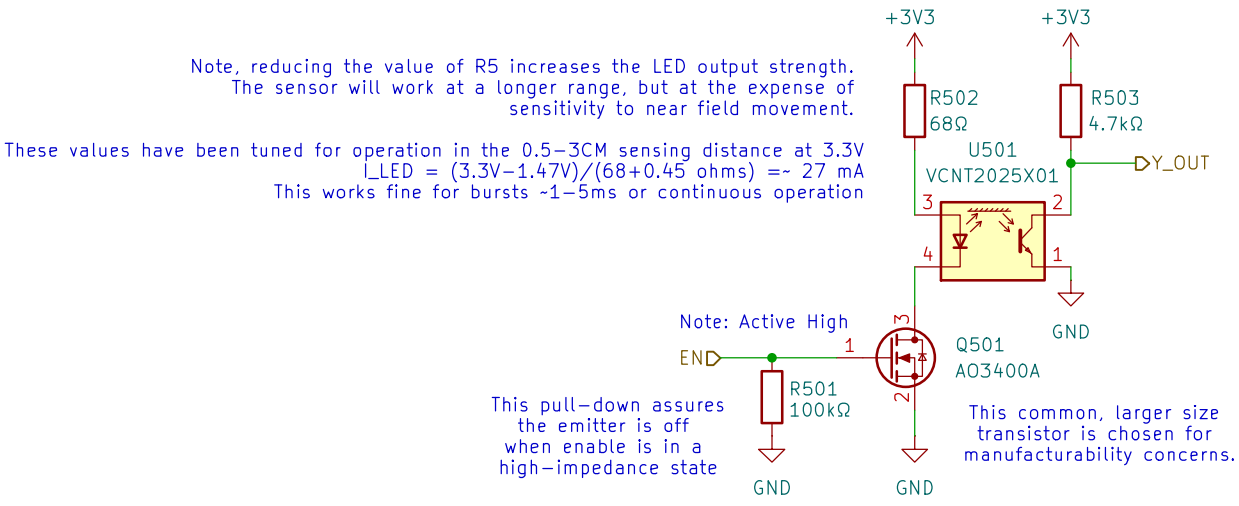
Size: A4

Rev: B

Id: 2/42







Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

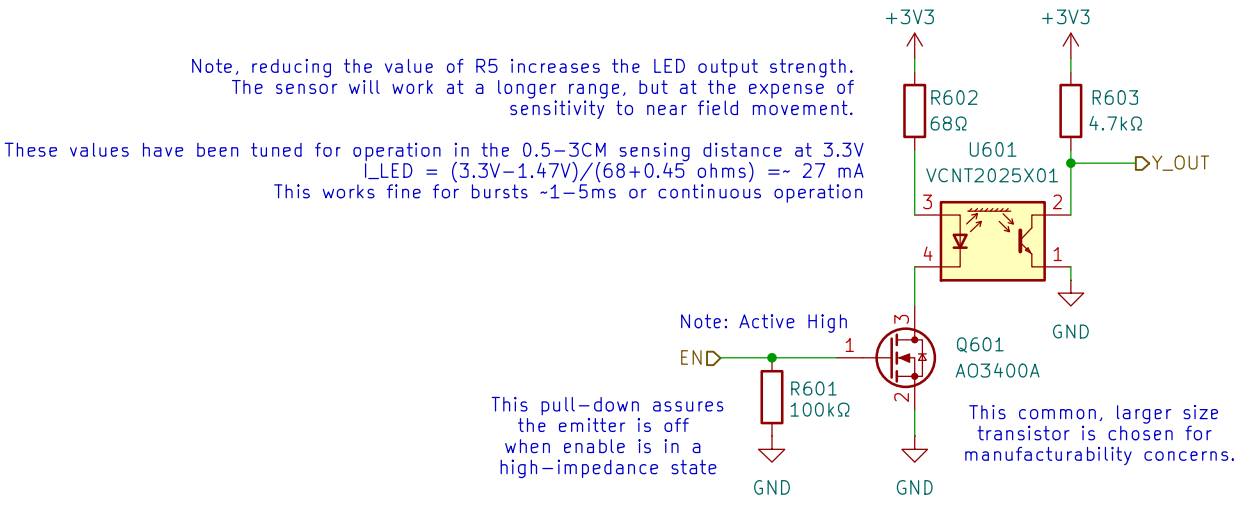
Size: A4

Date: 2026-04-22

Rev: B

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Id: 5/42



Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

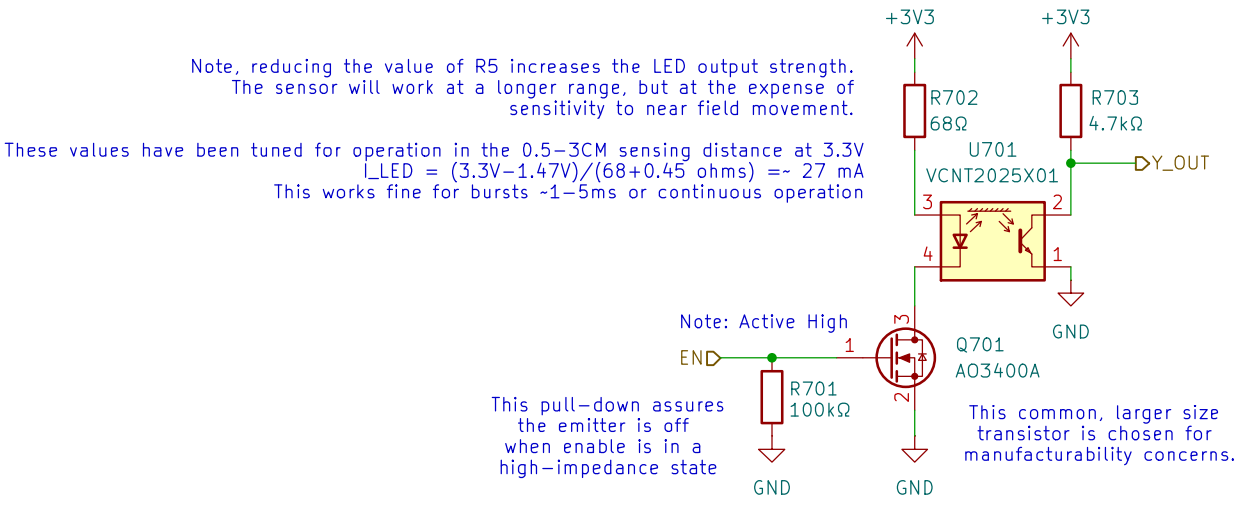
Size: A4

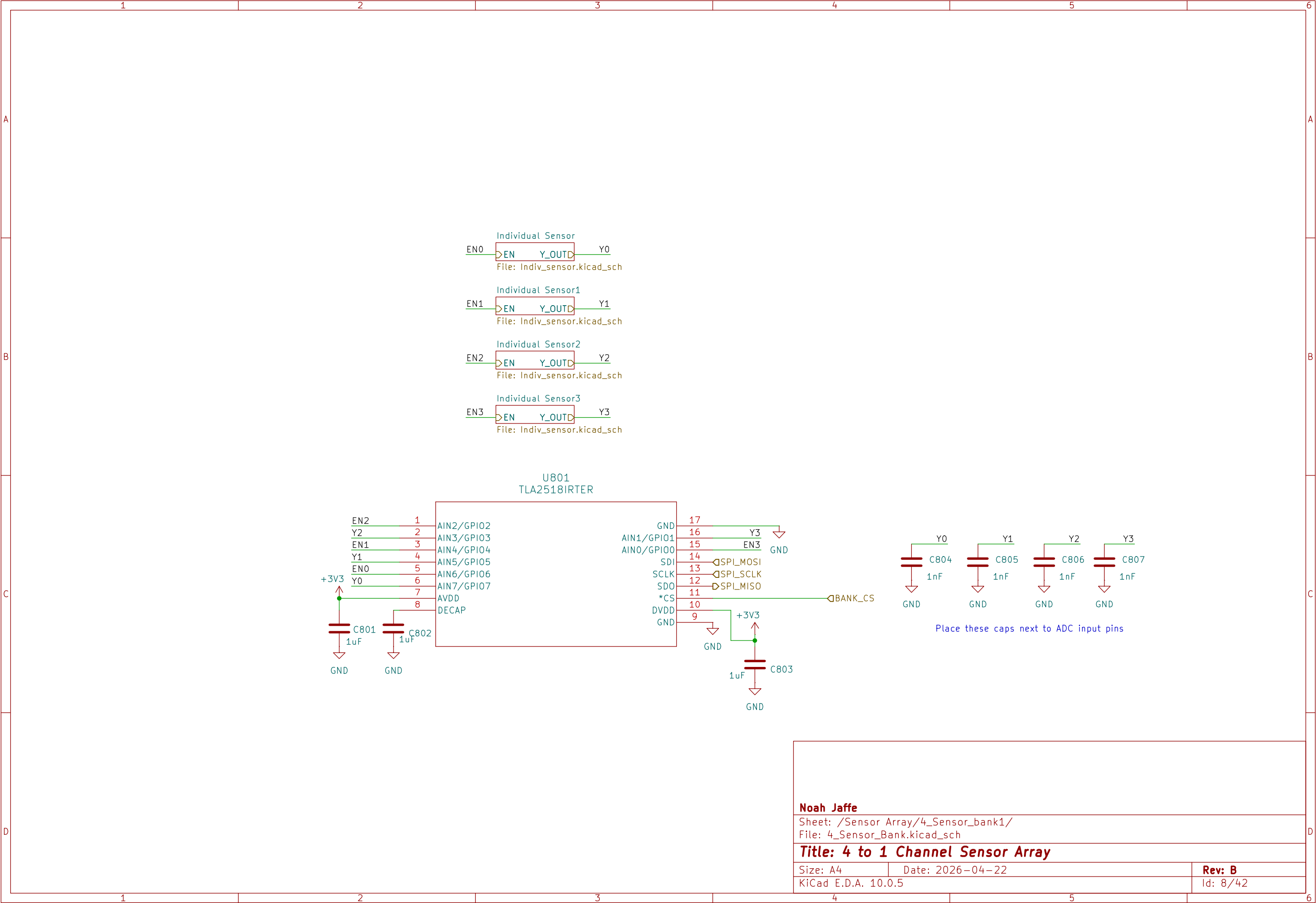
Date: 2026-04-22

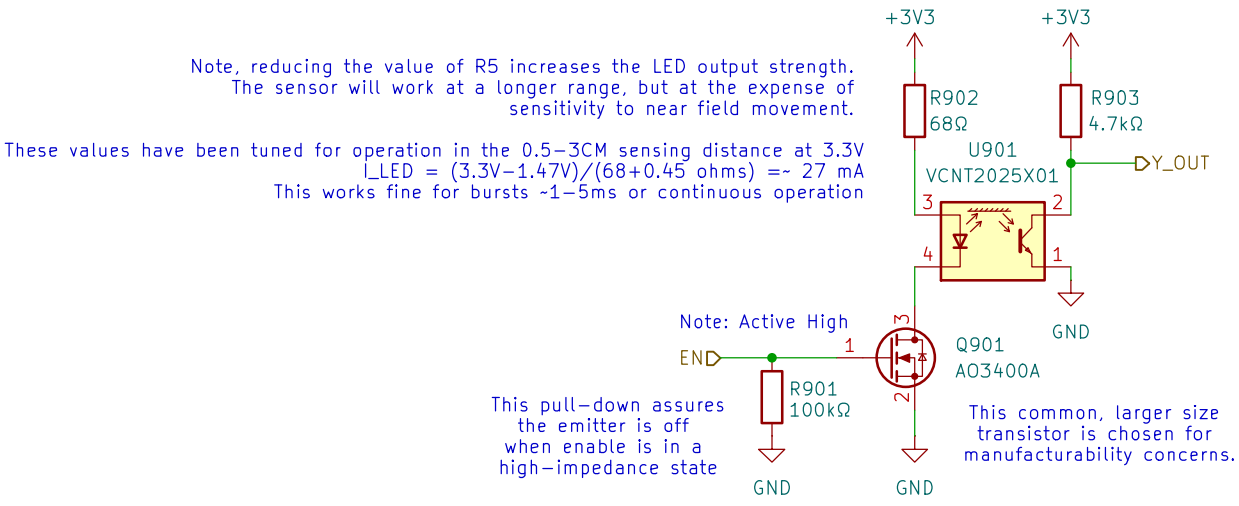
Rev: B

KiCad E.D.A. 10.0.5

Id: 6/42







Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank1/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

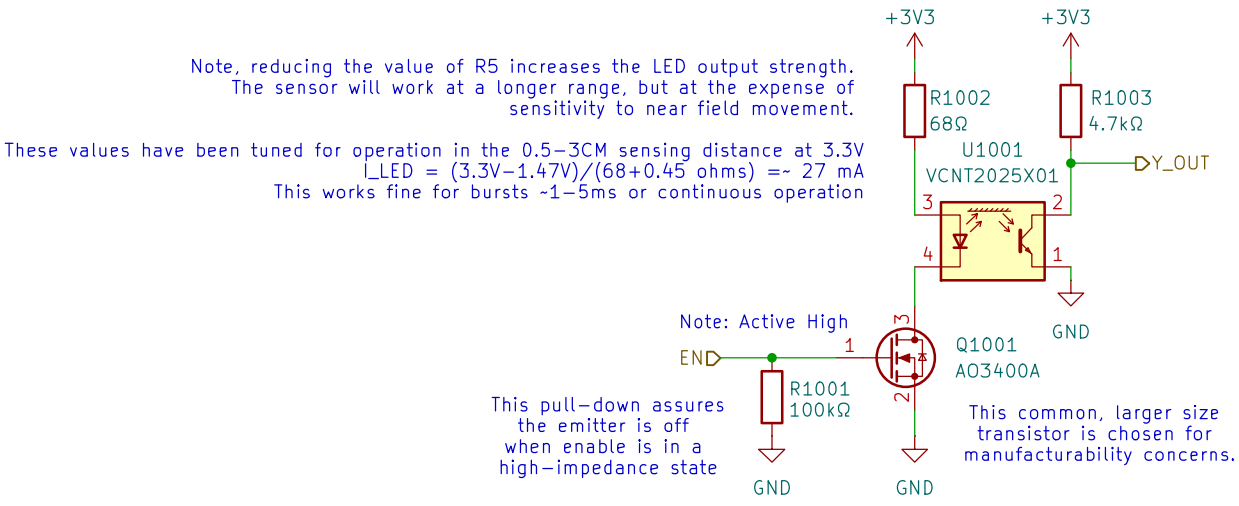
Size: A4

Date: 2026-04-22

Rev: B

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Id: 9/42



Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank1/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

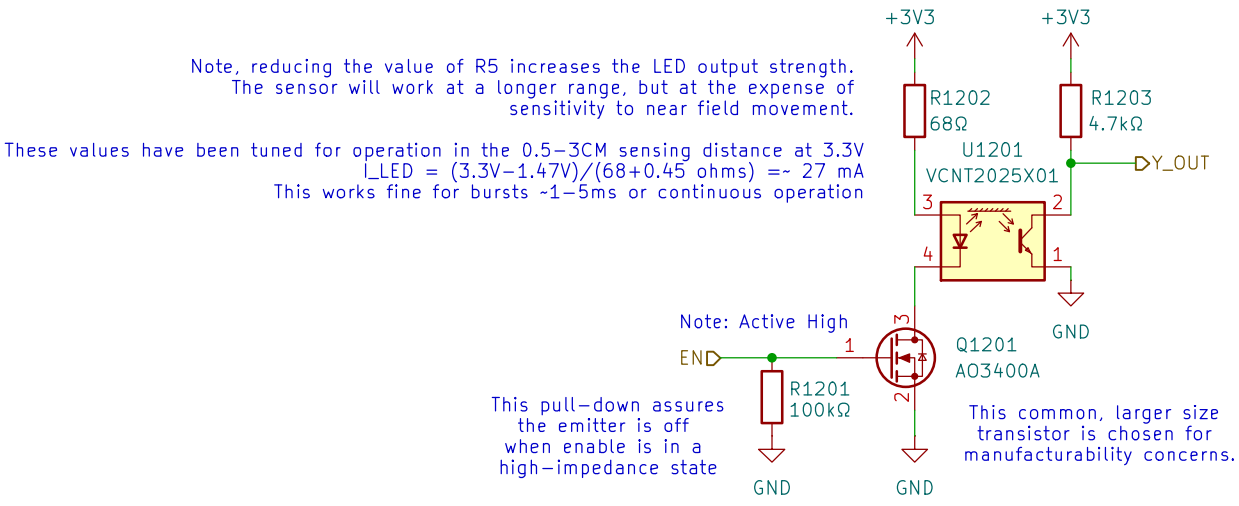
Size: A4

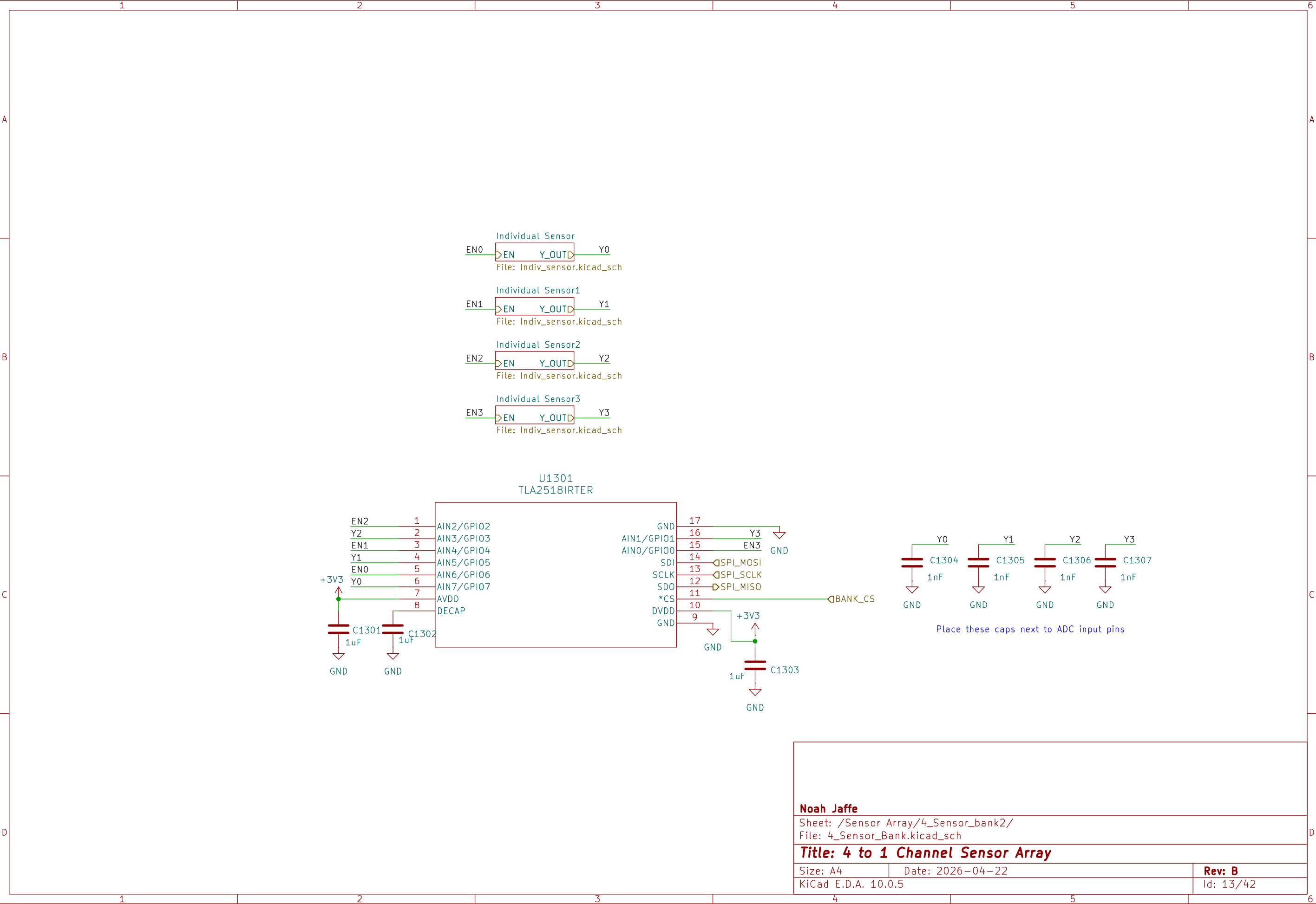
Date: 2026-04-22

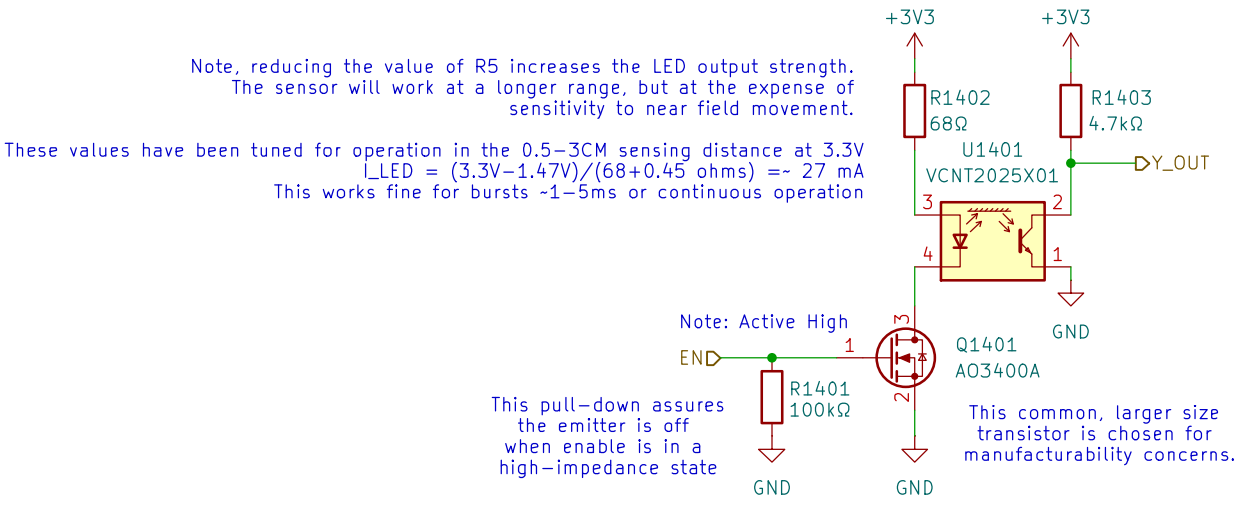
Rev: B

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Id: 10/42







Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

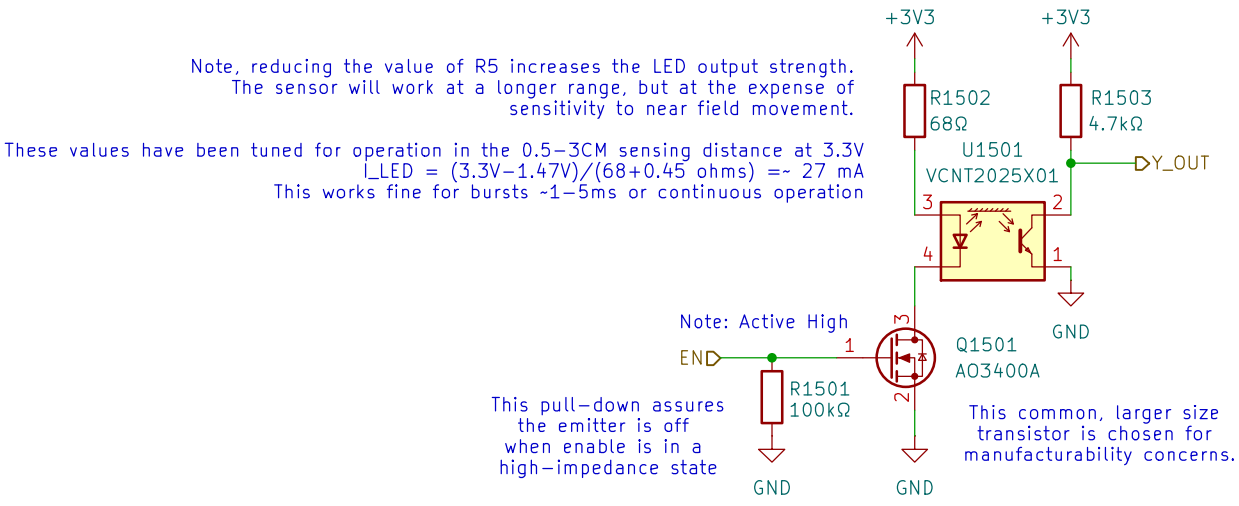
Size: A4

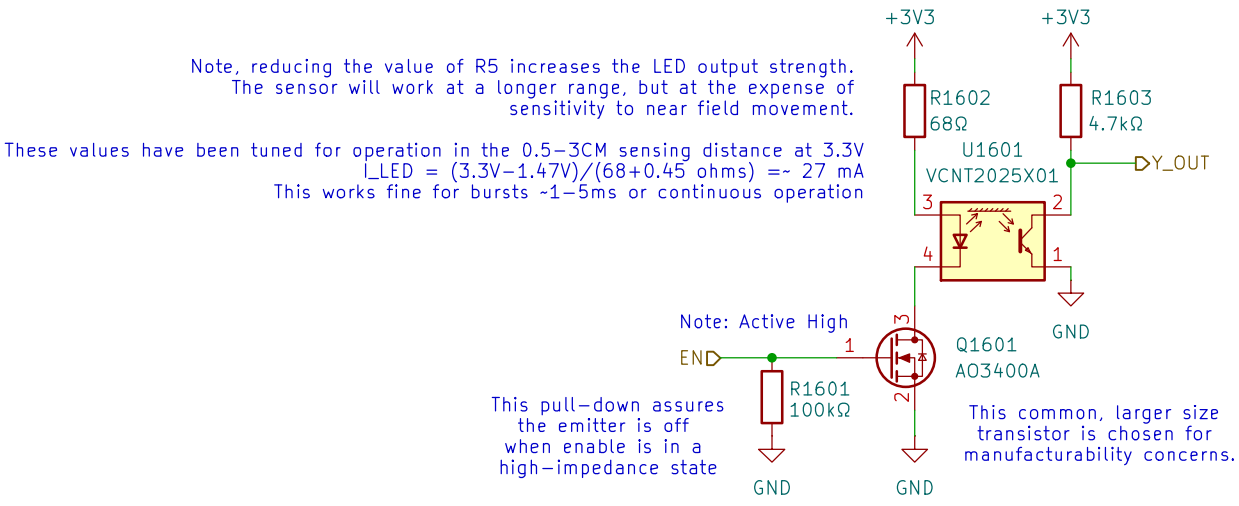
Date: 2026-04-22

Rev: B

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Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

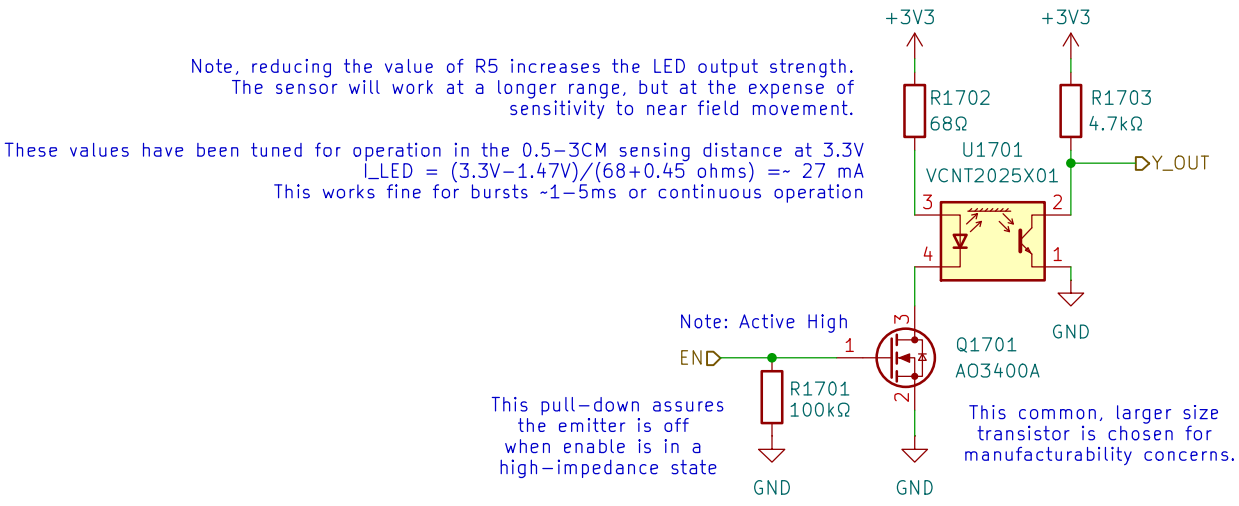
Size: A4

Date: 2026-04-22

Rev: B

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Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank2/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

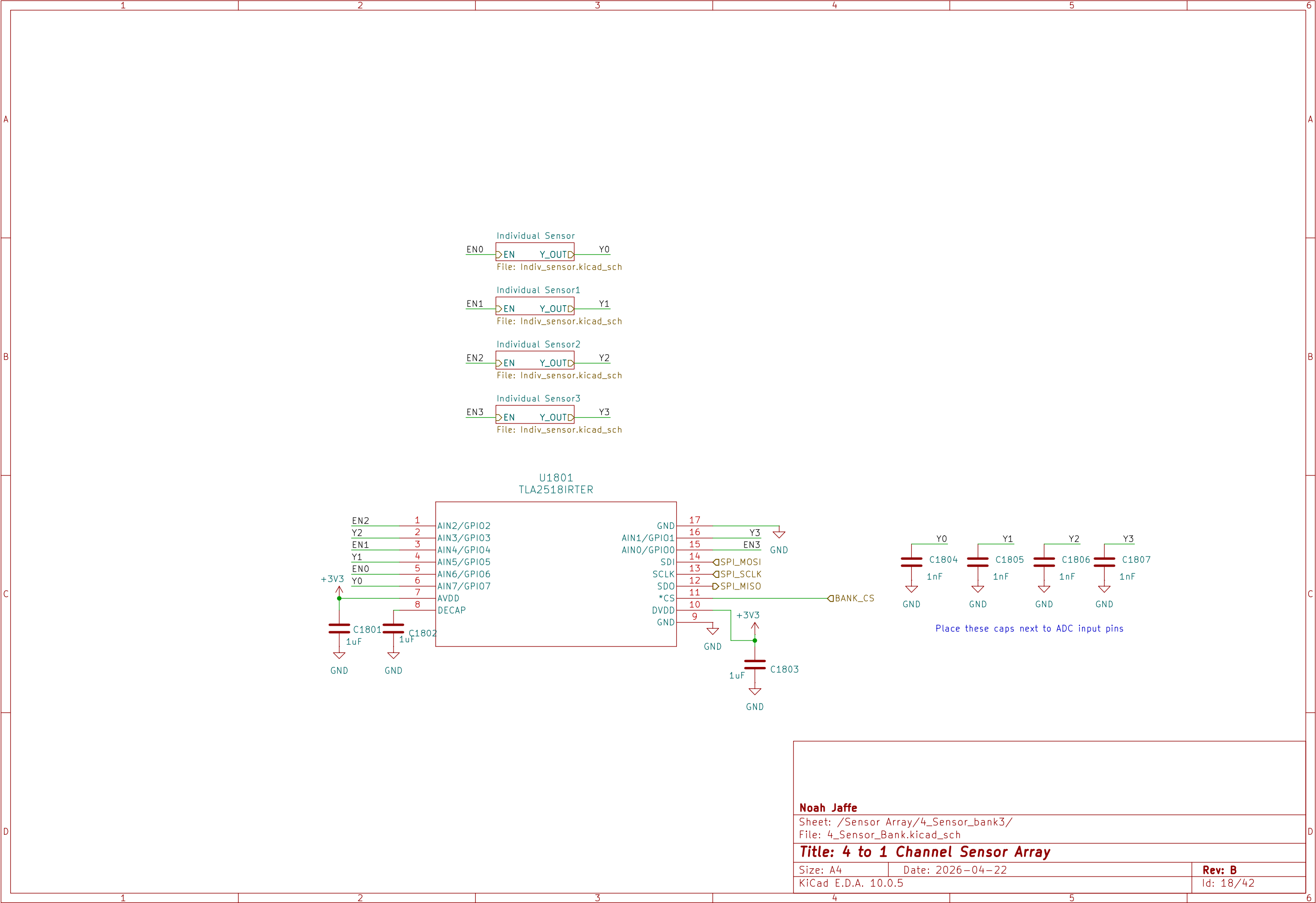
Size: A4

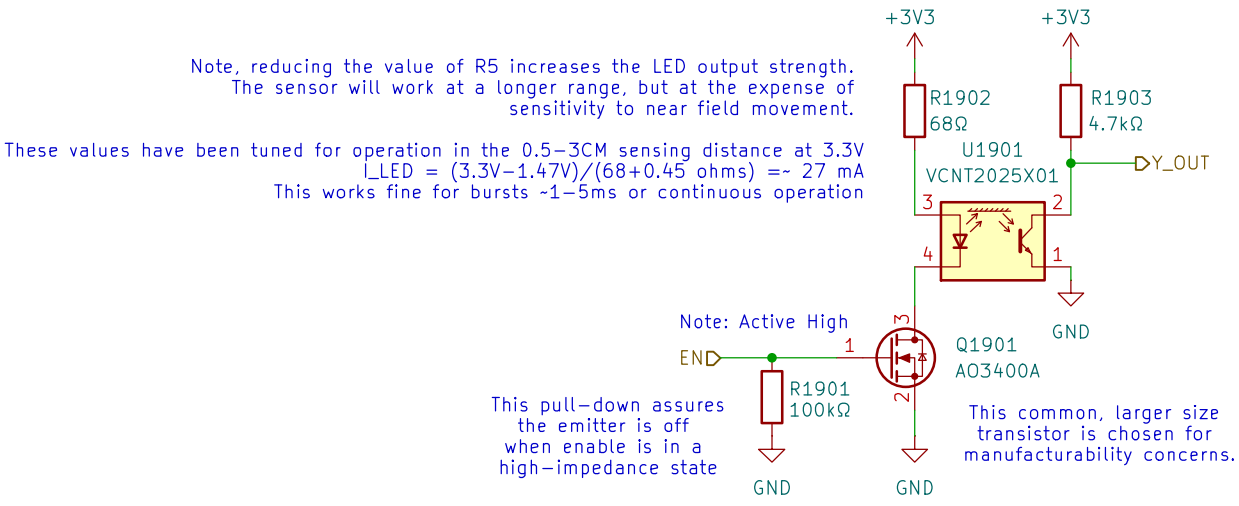
Date: 2026-04-22

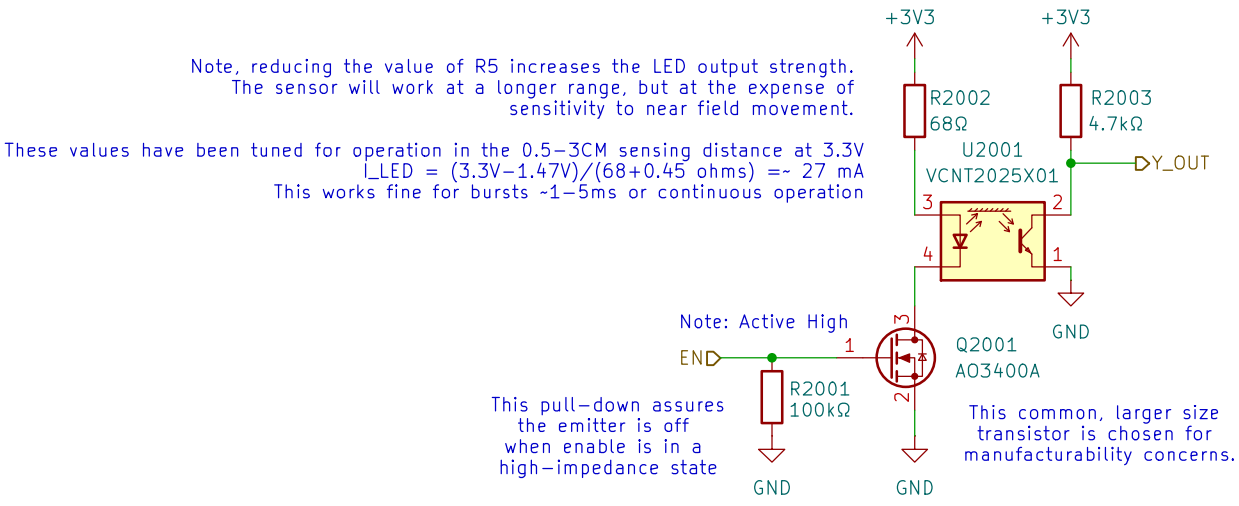
Rev: B

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Id: 17/42







Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

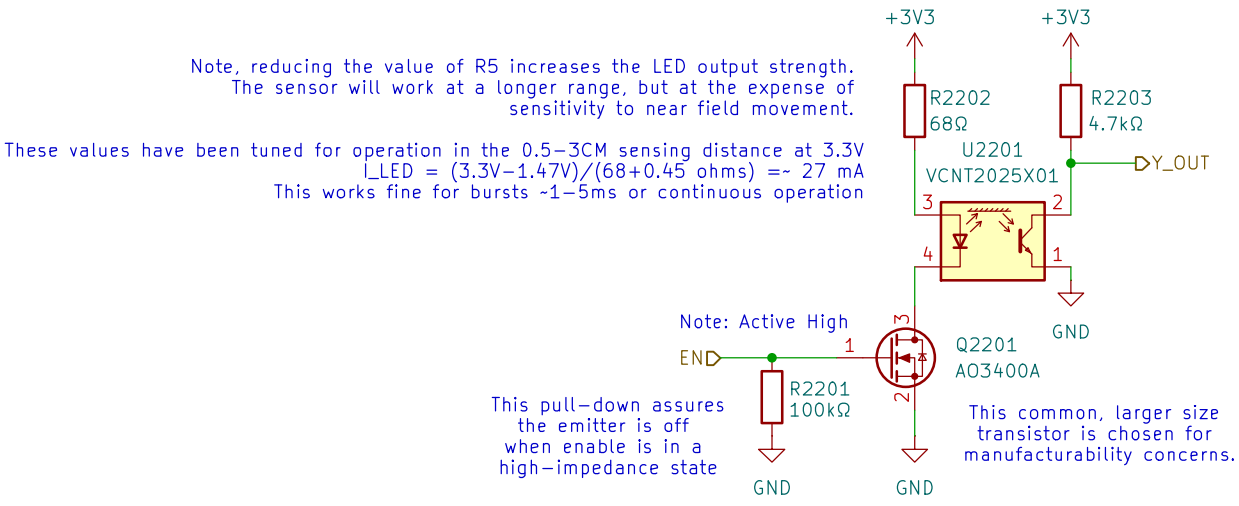
Size: A4

Date: 2026-04-22

Rev: B

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Id: 20/42



Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank3/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

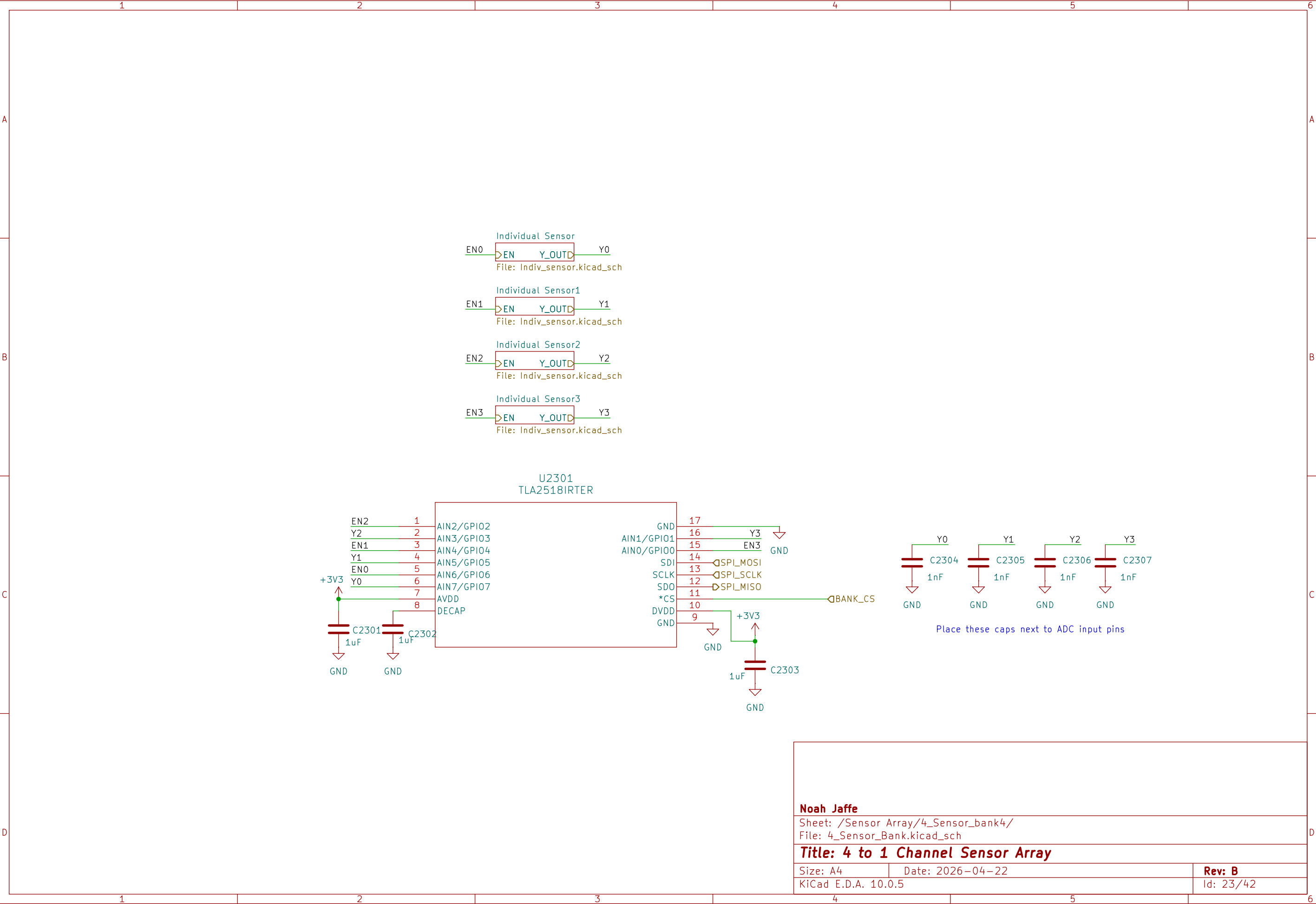
Size: A4

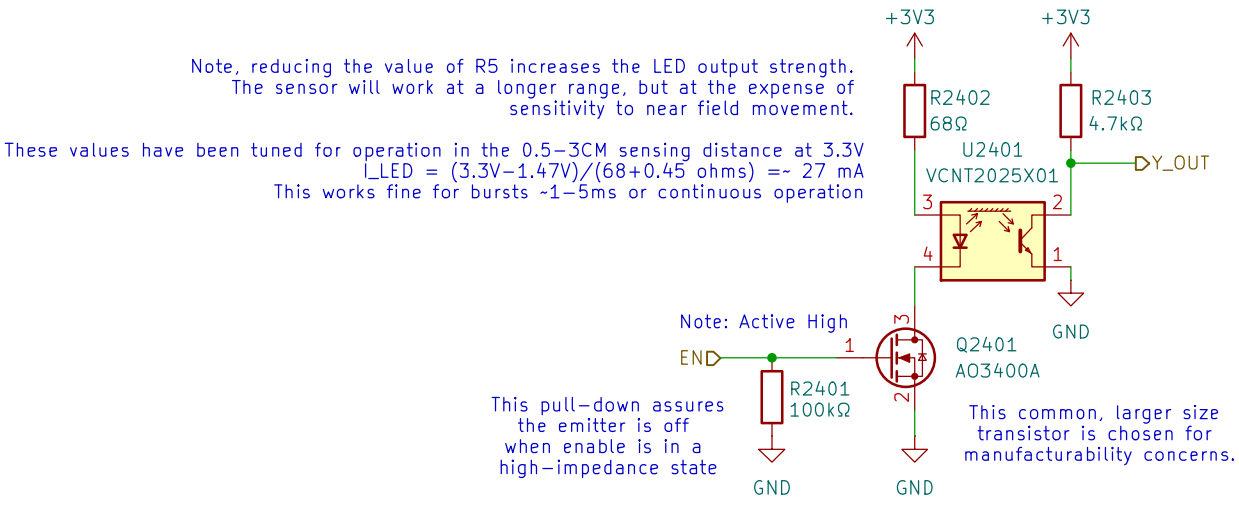
Date: 2026-04-22

Rev: B

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Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

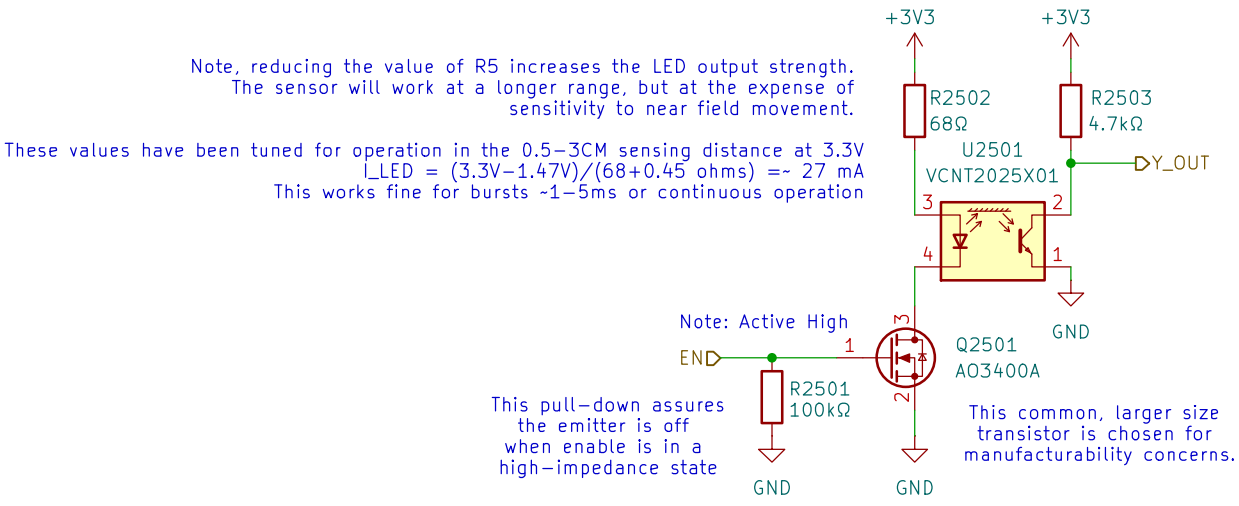
Size: A4

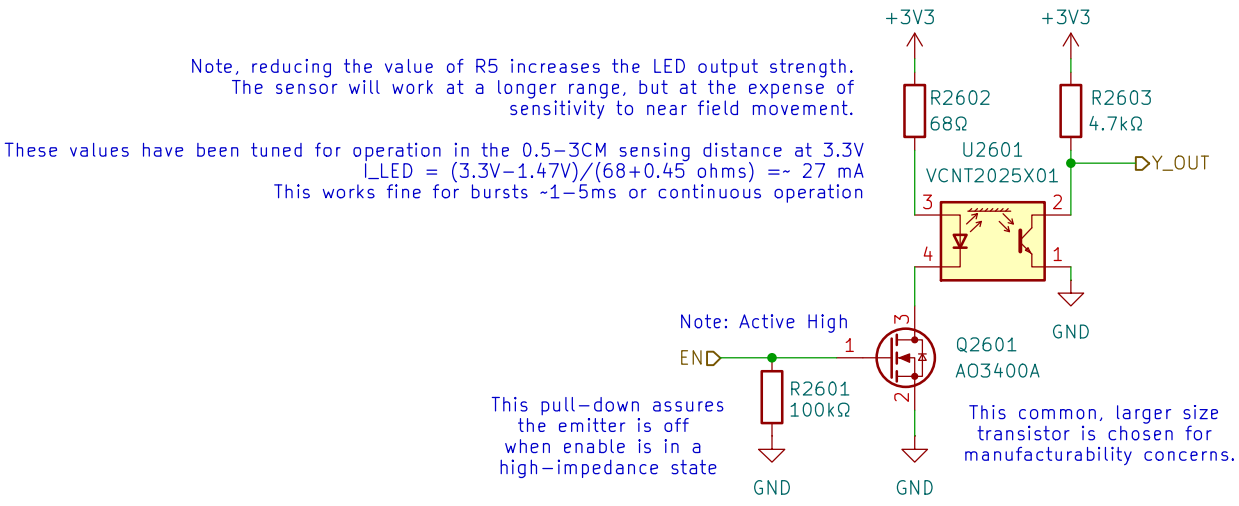
Date: 2026-04-22

Rev: B

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Id: 24/42





Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

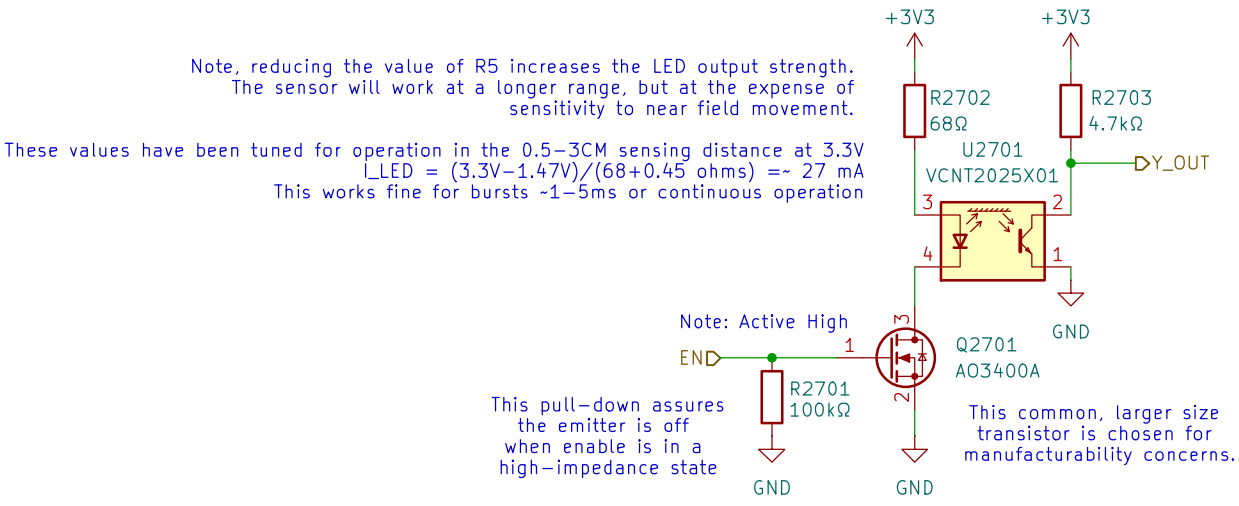
Size: A4

Date: 2026-04-22

Rev: B

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Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank4/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

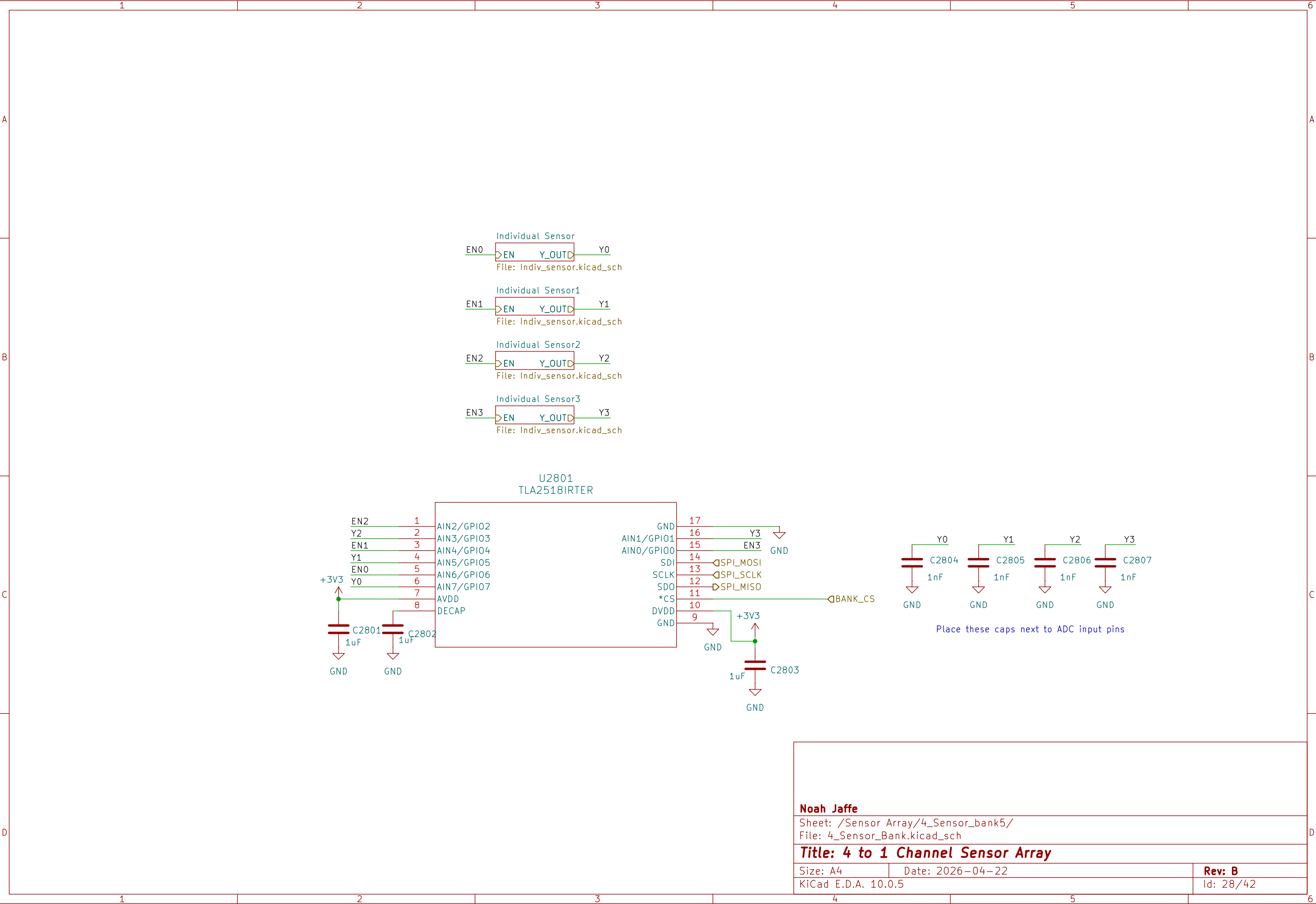
Size: A4

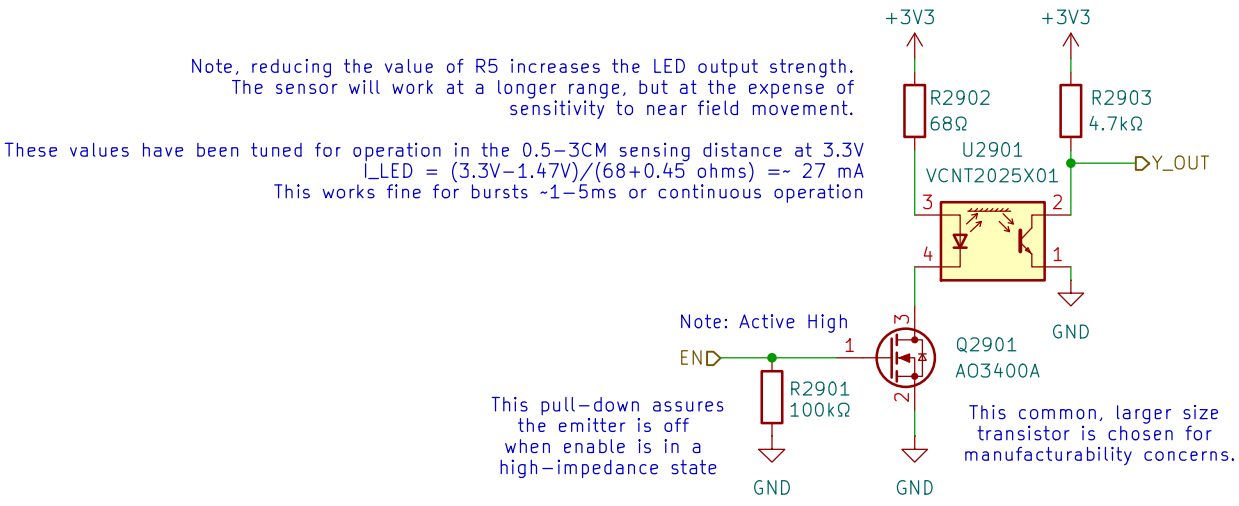
Date: 2026-04-22

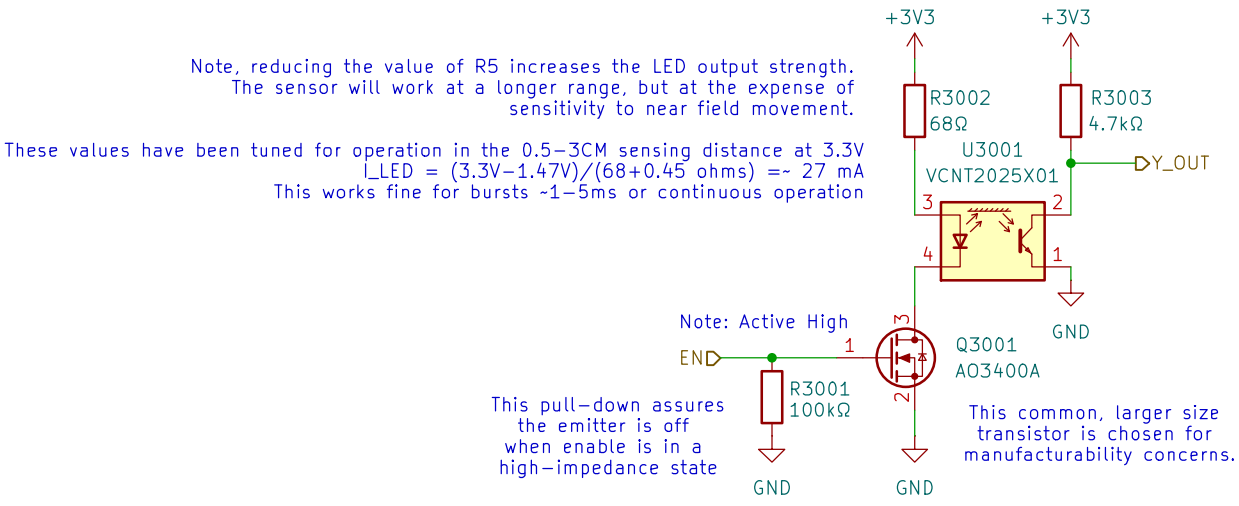
Rev: B

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Id: 27/42







Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor1/
File: Indiv_sensor.kicad_sch

Title:

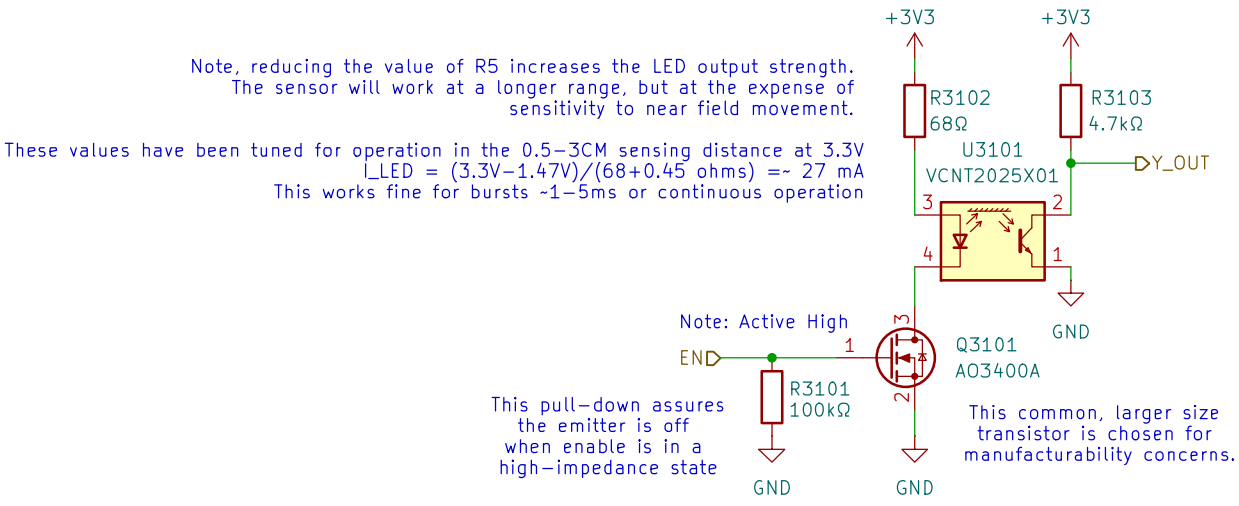
Size: A4

Date: 2026-04-22

Rev: B

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Id: 30/42



Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

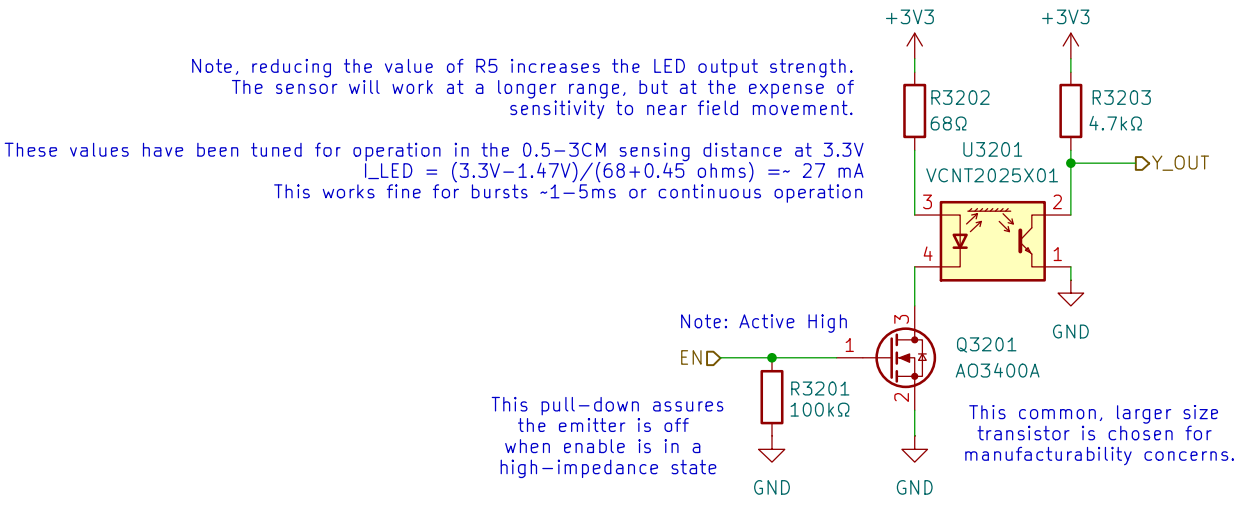
Size: A4

Date: 2026-04-22

Rev: B

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Id: 31/42



Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank5/Individual Sensor3/
File: Indiv_sensor.kicad_sch

Title:

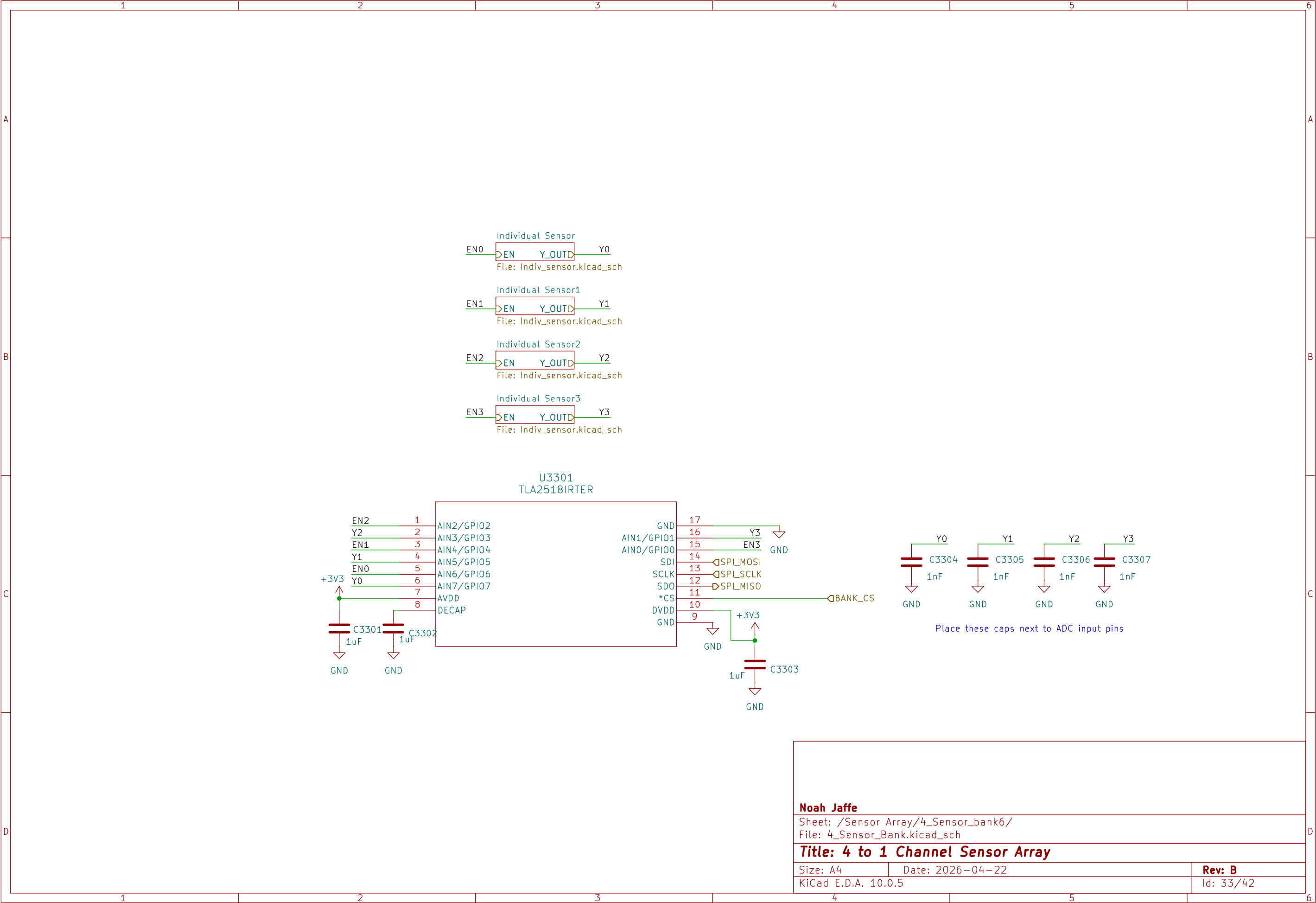
Size: A4

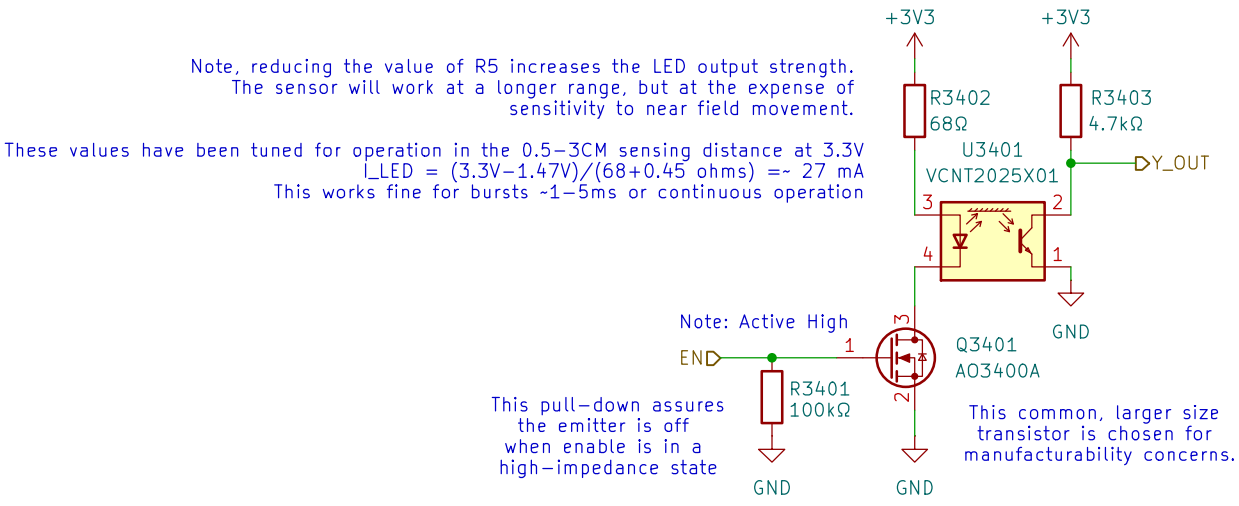
Date: 2026-04-22

Rev: B

KiCad E.D.A. 10.0.5

Id: 32/42





Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor/
File: Indiv_sensor.kicad_sch

Title:

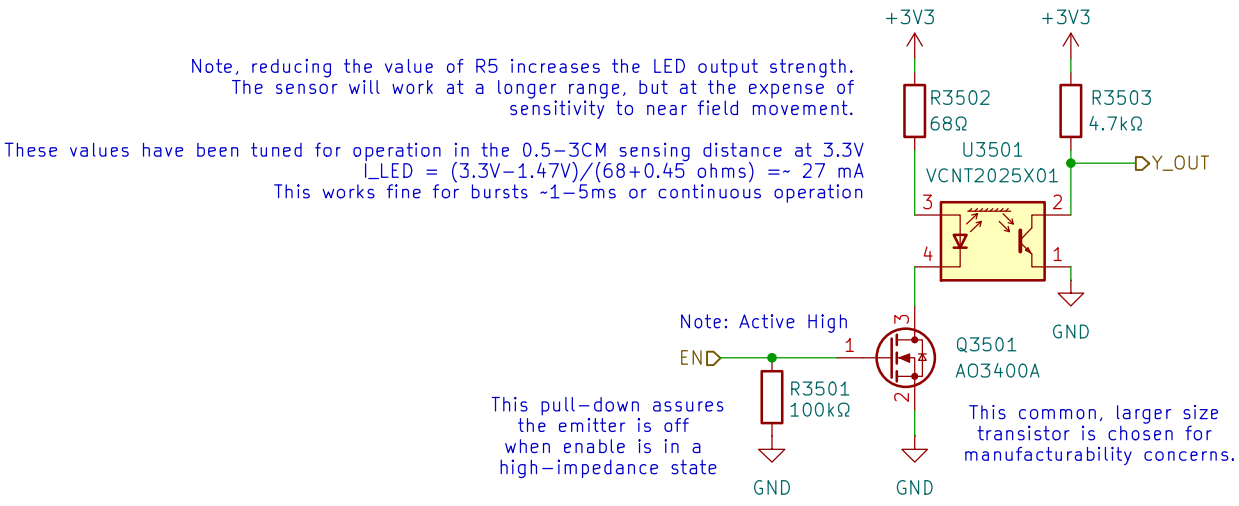
Size: A4

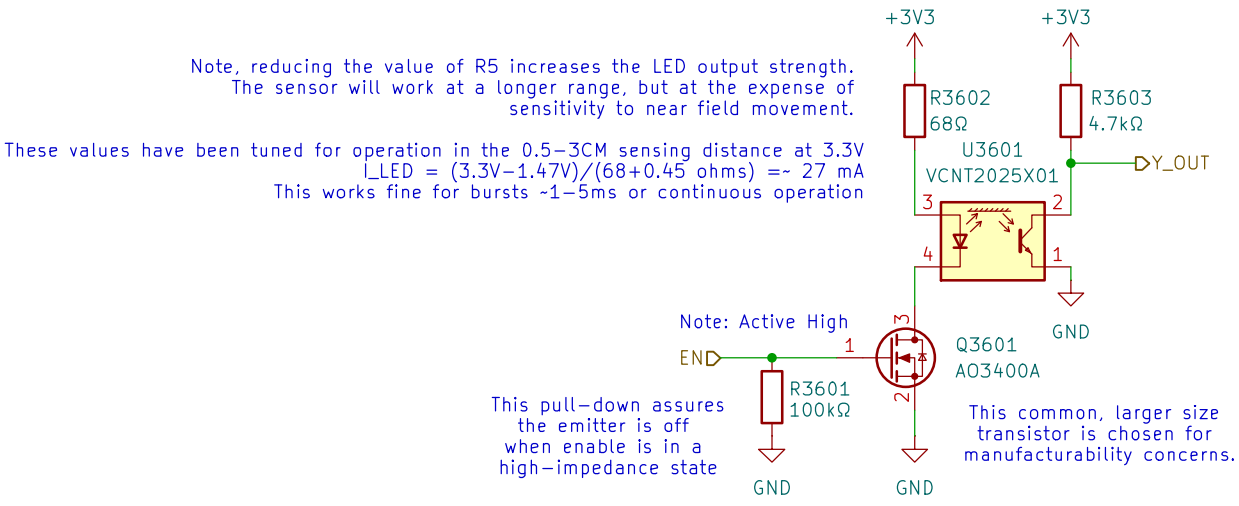
Date: 2026-04-22

Rev: B

KiCad E.D.A. 10.0.5

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Noah Jaffe

Sheet: /Sensor Array/4_Sensor_bank6/Individual Sensor2/
File: Indiv_sensor.kicad_sch

Title:

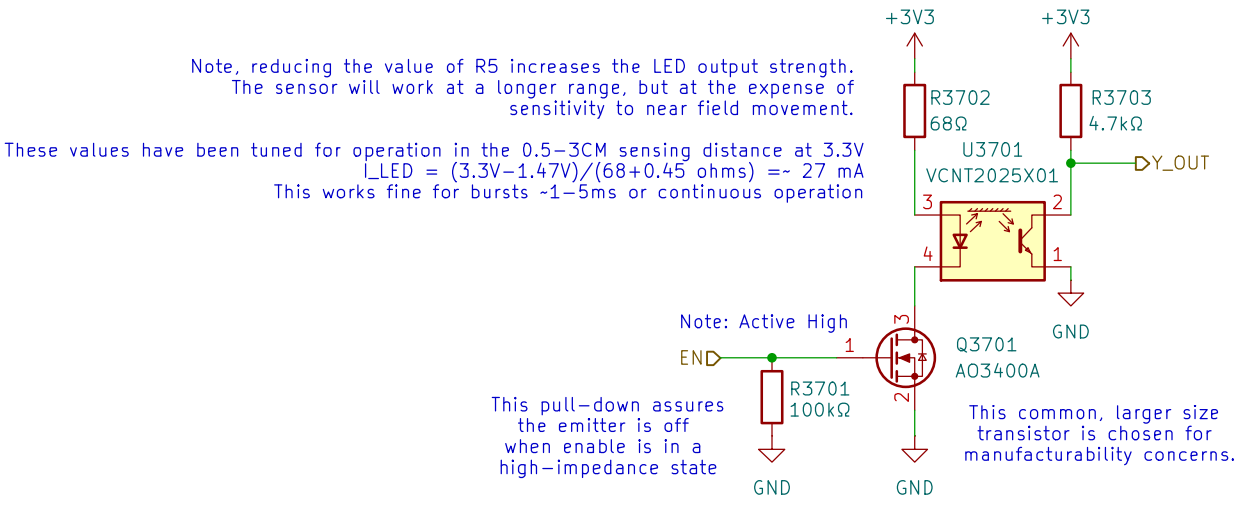
Size: A4

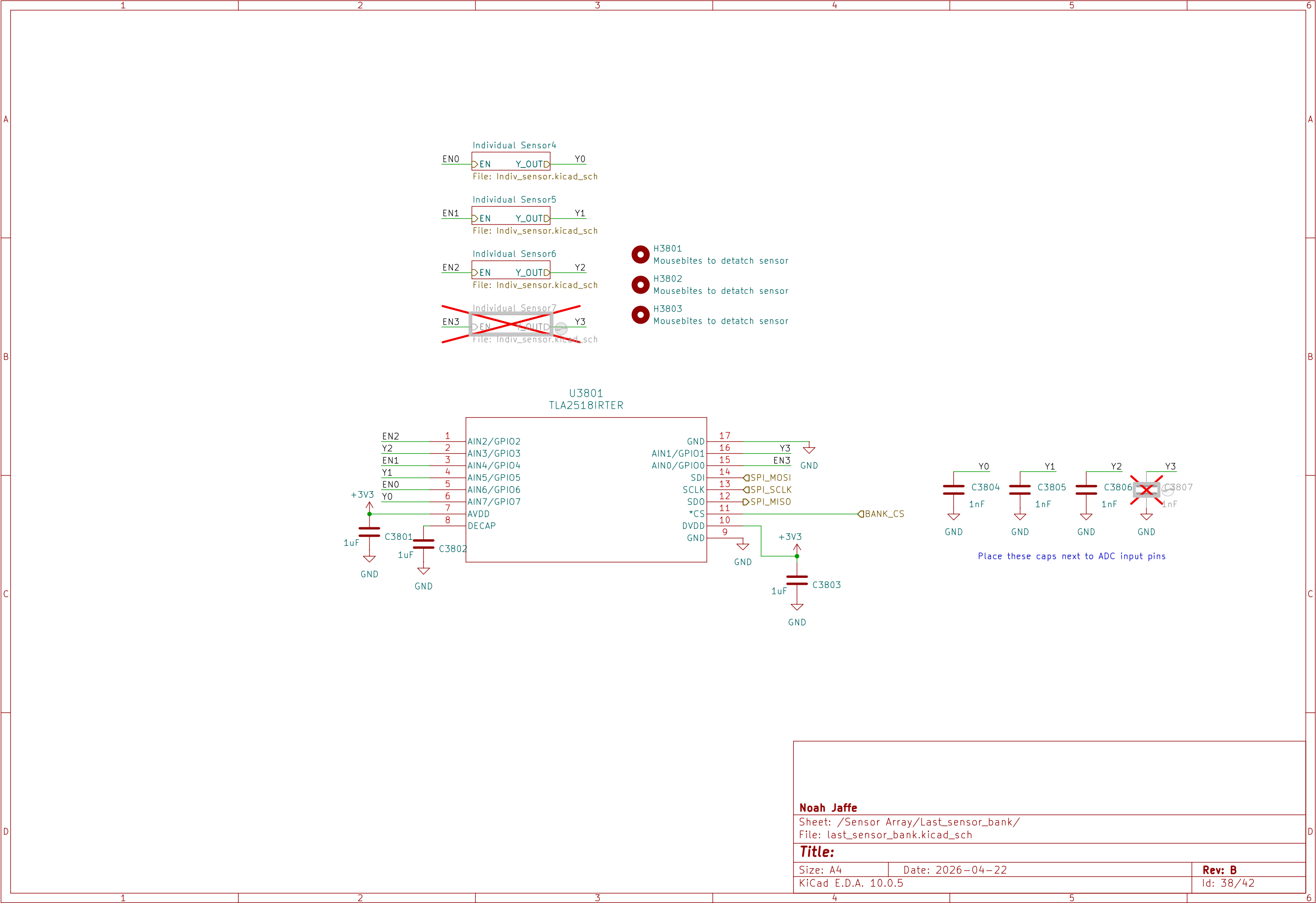
Date: 2026-04-22

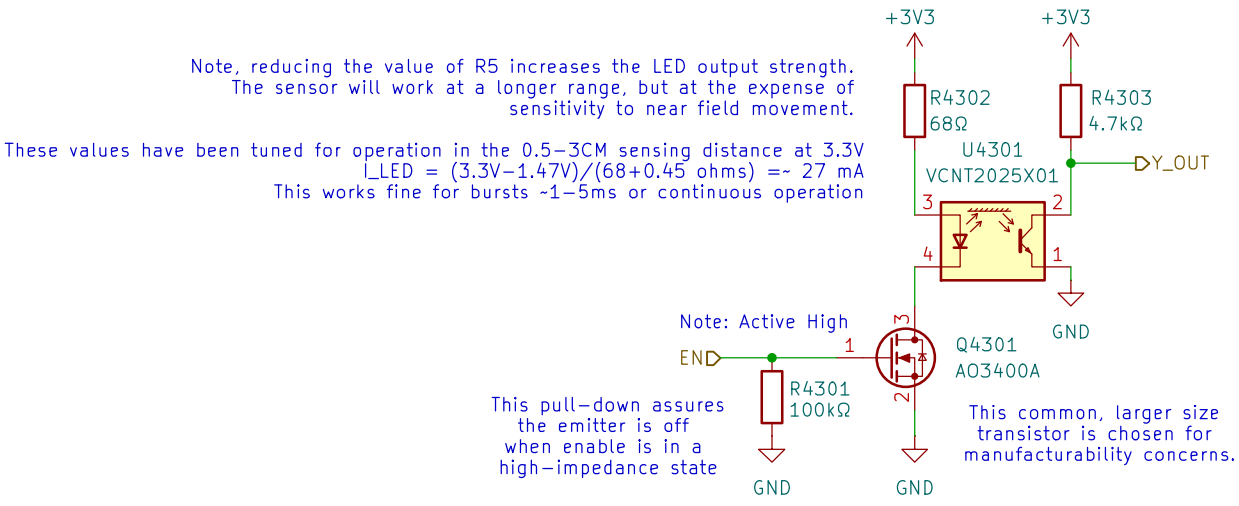
Rev: B

KiCad E.D.A. 10.0.5

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Noah Jaffe

Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor4/
File: Indiv_sensor.kicad_sch

Title:

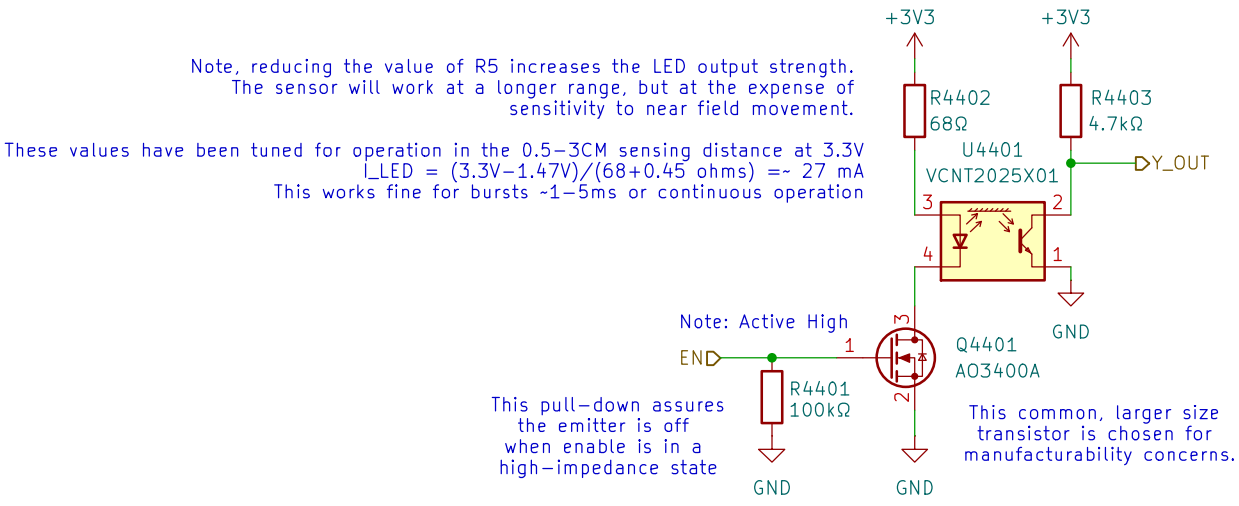
Size: A4

Date: 2026-04-22

Rev: B

KiCad E.D.A. 10.0.5

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Noah Jaffe

Sheet: /Sensor Array/Last_sensor_bank/Individual Sensor5/
File: Indiv_sensor.kicad_sch

Title:

Size: A4

Date: 2026-04-22

Rev: B

KiCad E.D.A. 10.0.5

Id: 44/42

